THE HOM COMPLEX OF A STRICTLY PERFECT COMPLEX INTO A K-FLAT COMPLEX IS AGAIN K-FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Computation of R-Hom out of a strictly perfect complex and the comparison map for tensor products. Stacks Project, Cohomology of Sheaves, Lemma lemma-Rhom-strictly-perfect (tag 0A8D) and Lemma lemma-diagonal-better (tag 0A8K). Supplies the description of the hom complex and the comparison map gamma used in the proof.
- K-flat complexes and acyclicity of tensor products. Stacks Project, Cohomology of Sheaves, Lemma lemma-local-actual (tag 0A8L). Gives the acyclicity of the total tensor product with a K-flat complex.

Source reference. Original lemma 08DM: sources/stacks-chow/tags.txt, source lines 10642–10642. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BYR: sources/stacks-chow/tags.txt, source lines 15024–15024. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08C9: sources/stacks-chow/tags.txt, source lines 10594–10594. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0118: sources/stacks-chow/tags.txt, source lines 1354–1354. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In the situation of Lemma 08DM if $\mathcal{F}^\bullet$ is K-flat, then $\mathcal{H}^\bullet$ is K-flat.*

Proof. Observe that $\mathcal{H}^\bullet$ is simply the hom complex $\mathcal{H}om^\bullet(\mathcal{E}^\bullet, \mathcal{F}^\bullet)$ since the boundedness of the strictly prefect complex $\mathcal{E}^\bullet$ insures that the products in the definition of the hom complex turn into direct sums. Let $\mathcal{K}^\bullet$ be an acyclic complex of $\mathcal{O}_X$-modules. Consider the map

$$\gamma : \mathrm{Tot}(\mathcal{K}^\bullet \otimes \mathcal{H}om^\bullet(\mathcal{E}^\bullet, \mathcal{F}^\bullet)) \longrightarrow \mathcal{H}om^\bullet(\mathcal{E}^\bullet, \mathrm{Tot}(\mathcal{K}^\bullet \otimes \mathcal{F}^\bullet))$$

of Lemma 0BYR. Since $\mathcal{F}^\bullet$ is K-flat, the complex $\mathrm{Tot}(\mathcal{K}^\bullet \otimes \mathcal{F}^\bullet)$ is acyclic, and hence by Lemma 08C9 (or Lemma 08DM if you like) the target of γ is acyclic too. Hence to prove the lemma it suffices to show that γ is an isomorphism of complexes. To see this, we may argue by induction on the length of the complex $\mathcal{E}^\bullet$. If the

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length is ≤ 1 then the $\mathcal{E}^\bullet$ is a direct summand of $\mathcal{O}_X^{\oplus n}[k]$ for some $n \geq 0$ and $k \in \mathbf{Z}$ and in this case the result follows by inspection. If the length is > 1, then we reduce to smaller length by considering the termwise split short exact sequence of complexes

$$0 \to \sigma_{\geq a+1}\mathcal{E}^\bullet \to \mathcal{E}^\bullet \to \sigma_{\leq a}\mathcal{E}^\bullet \to 0$$

for a suitable $a \in \mathbf{Z}$, see Homology, Section 0118. Then γ fits into a morphism of termwise split short exact sequences of complexes. By induction γ is an isomorphism for $\sigma_{\geq a+1}\mathcal{E}^\bullet$ and $\sigma_{\leq a}\mathcal{E}^\bullet$ and hence the result for $\mathcal{E}^\bullet$ follows. Some details omitted. $\square$